

System Specific Information

The table below lists configuration information about the system which may be system specific. This table should be used as a reference when reviewing system specifications and capabilities found in the "Hardware Reference Manual".

Consult the "Hardware Reference Manual" for all system specifications.

General	
System	Redstone HF-2
Serial Number	38184
Hardware	
F1 Channel	
Freq Range	1 MHz - 600 MHz
Phase/Amplitude Mod Type	EM-III
Freq Control Type	EM-III
F2 Channel	
Freq Range	0.5 MHz - 500 MHz
Phase/Amplitude Mod Type	EM-III
Freq Control Type	EM-III
Gradients	
Gradient Type	None
Rotation Option	
B_0 Compensation	

Phase Reset

In order to ensure phase synchronization between all transmitters and receivers, all pulse sequences need to begin with a recommended 5 μs event (minimum duration) for the following sequence lines:

F0_PhRst F1_PhRst

RX_PhRst F2_PhRst

Pulse Sequence Control Lines:

Transmitter:

FN_Ampl – Linear control of the transmitter output. Valid values are 0 – 100

FN_Atten – Attenuates the transmitter output. Valid values are 0 - 63.5 dB in 0.5 dB steps.

FN_Phmod – Fine phase control of the transmitter output.

FN_Ph – Two bit phase control of the transmitter output. Valid Values are 0, 90, 180, 270 degrees, or 0,1,2,3 if step is used.

FN _Unblank – TTL output meant to unblank the amplifier.

FN _TXGate – This gates the transmitter output.

FN_Freq – Allows the transmitter or LO frequency to be changed within the pulse sequence. The frequency change persists into the following events until the frequency change is set to 0. The value needs to be in Hz.

Acquisition:

Acq – Event in which the acquisition begins.

Acq_Phase – Two bit phase control of the receiver. Valid values are 0, 90, 180, 270 degrees or
0,1,2,3 if step is used.

RX_Phase – Fine phase control of the reciever

RX_Blank – Blanks the receiver.

Extra:

Scope_Trigger – TTL output that can be used to trigger external devices like an Oscilloscope.

General

Some important limitations for Redstone/EM-III systems are summarized here:

Description	Value	Notes
Minimum Event Duration	100 ns 200 ns	Shorter pulses (10ns min) can be created using a waveform. Special case for: 1) event before external trigger 2) event before waveform.
Maximum Event Duration	42.94967295 s	May be exceeded for some event types – compiler will create additional duplicate events
Minimum Events in Sequence	2	Generally requires 3 events if an acquisition is present
Maximum Events in Sequence	3071 – (# of loops) – (# of extended wfms)	
Maximum Tables in Sequence	~256 8·(# tables) + (# non-waveform entries) 910	See detailed discussion below With shared memory option.
Maximum Half-D Tables in Sequence	512	
Maximum Loops in Sequence	680	
Maximum Waveform Duration	42.949s · (# samples)	

Table space calculation in the EM-III is somewhat complex. Table memory (nD tables & Waveform headers) is at 0xC0000-0xC1FFF. Here are the rules:

- Headers are 8 words each.
- Any ND table automatically creates any missing lower-dimensional tables for the same event. A half-D table creates 1D-4D.
- Space for each table is allocated based on the # of entries for that table rounded up to the nearest multiple of 4.
- Waveform sample points are not stored in the table memory, but in the waveform memory (SO-DIMM), so they do not count against the table memory space.
- If the shared memory option is in use, table entries are also stored in waveform memory, but the table header requirement increases to 9 words each.

Example: Event 3: 2D table with 5 entries; Event 5: Waveform table with 128 entries; Event 7: half-D table (17 entries) with 2D modifier (4 entries). This occupies 112 words of table memory, and 128 words of waveform memory. Calculate this as follows: Event 3, 2D Header + 8 entries (5, rounded-up) + 1D Header + 4 entries (1 default, rounded-up); Event 5, Waveform header; Event 7, half-D header + 20 entries (17, rounded-up) + 4D header + 4 entries (1 default, rounded-up) + 3D header + 4 entries (1 default, rounded-up) + 2D header + 4 entries + 1D header + 4 entries (1 default, rounded-up). Total 8 headers + 48 entries = 112 words. Note that half-D tables are treated as "5D" for this calculation.

Generalization: If the average table dimension and size is 2D, with 12 entries, then the average space required is 16 (2 headers) + (12 + 4 entries), so there is room for about 256 of these.

External Trigger

- The duration of the external trigger event is ignored – the event may be shorter *OR* longer than specified in the sequence.
- No external trigger allowed on the first event (or last event?) of sequence.
- External trigger signal interval (between consecutive triggers) must be > 20μs.
- The event before the external trigger event must have a duration > 200ns.

Waveform Tables

Some rules:

- Event before a waveform must be at least 200n
- Minimum waveform is 100n
- Minimum time/pt is 10n
- Waveforms must have at least 2 samples.
- Back-to-back waveforms are allowed (but first waveform must then be at least 200n long)
- No waveform on an end loop, condition test, or conditional branch event
- "Long waveforms" are allowed (> 110404). These waveforms can arise if a waveform table is continued over a number of other events whose total time exceeds the single-event limit (~42.9s).
- Continued waveforms may not cross into or out of a loop.
- Conditional branches in the main sequence may not occur into, out of, or during, continued tables.
- A waveform in a loop must have an even number of points, if shared waveform memory is in use, otherwise the waveform will lose synchronization with the main sequence.

Processing of "mismatched" waveforms.
These are waveforms for which (dwell time) • (number of samples) ≠ (event duration). This condition can occur if the user enters a smaller dwell time than the compiler would generate normally, or if the event duration is not divisible by an integer number of samples. In either case, the waveform data will be exhausted before the end of the event. There are two possibilities: 1) stop at end of table; or 2) recycle, starting over from top of table. The option in use is controlled by bit 11 of the ConfigWord (1: recycle; 0: stop - default). Special rules:

- Mismatched waveforms to be recycled must contain an even number of samples (divisible by 2 if dwell > 10n, divisible by 4 if dwell = 10n).
- After "stop", the last entry is retained through the end of the waveform event.

Variable-duration waveforms
The presence of timer (delay) tables in the sequence may cause the waveform duration to change. In this case, the waveform will always be "mismatched". If this usage is intended, the "recycle" option should be enabled, and the time per point for the waveform should be specified in the table dialog window. The principal application of this feature is for decoupling schemes that run during variable preparation or evolution times in a 2D/nD experiment.

Unpaired amplitude/phase waveforms
As noted elsewhere, the Redstone uses a single event module to control both amplitude and phase waveforms for a single synthesizer. When two waveform tables are paired in the sequence editor, they should have the same number of samples (and the same time/pt). If the tables do not match, a

warning is given. By default, the shorter waveforms will be stretched to match the longest one in the same event. INI file option `StretchUnpairedWfm = 0` is provided in case the waveform should be repeated instead. (1/26/2015). The time per sample will be calculated based on the longest waveform (or use the time/pt value from its table).

Waveforms and loops

The following table indicates allowable combinations of waveforms (or continued waveforms) and loop events. Several cases should be fixed in future software/firmware releases.

Case	Events before loop	LL event	Events during loop	EL event	Events after loop	EM-III Std	EM-III Shared Memory Option
1	■					**OK**	**OK**
2		■				**OK**. (1, 2)	**OK**. (1, 2, 3)
3			■			**OK**	**OK**. (4)
4				■		**OK**	(5)
5					■	**OK**. (6)	OK (6)
6		■	■			**OK**. (1, 2)	**OK**. (1, 2, 3)
7			■	■		(7, 10)	(7, 10)
8		■	■	■		(7, 10)	(7, 10)
9	■	■	■	■		**OK**. (8)	**OK**. (8)
10		■	■	■	■	(7, 10)	(7, 10)
11	■	■	■	■	■	**OK**. (8, 9)	**OK**. (8, 9)
12	■	■	■			(7)	(7)
13			■	■	■	(7)	(7)

NOTES:	
■	Indicates waveform is present in this (these) event(s)
1	Not tested for variable loop duration.
2	Fails for loop tables.
3	Needs one non-waveform event inside loop
4	Needs one event between end of waveform and end-of-loop
5	Fails without warning message.
6	Possible timing issues if half-d tables are present in the loop.
7	Prohibited by compiler
8	Loop is eliminated by the compiler
9	May not work right if loop length is variable (delay tables or loop tables).
10	Should be allowed in future release

Loops and "Half-D" Tables

General rules:

- No waveform on loop end
- No waveform on loop start, if a loop table is used.
- With half-D tables, the event before the loop end must be long enough to update the tables (typically ~1μs/table)

The precise limits on timing for half-D table updating are complicated, depending upon:

- Total number of half-D tables in the loop
- Whether a half-D table is present on the first event
- Total number of events in the loop
- Duration of the last event.

Remember that the tables in each event are counted as separate tables, *even if they have the same name and contents*. If timing is critical, correct performance should be verified with an oscilloscope or loopback experiment before attempting an experiment with high power.

Asynchronous Subsequences

The timing of the subsequence and the main sequence must match exactly at all times. The compiler/assembler will attempt to fix up the composite sequence to comply with this, by adding events and/or delay tables. However, not all cases can be handled at this time.

General rules:

- When writing a subsequence, ignore the 1μs last event (which cannot be edited in the sequence editor). It will be discarded when the subsequence is embedded into the main sequence.
- A subsequence must have at least two events (not counting the 1us last event).
- Only simple events and nD tables are allowed in the subsequence.
- Avoid conditional branches in the main sequence into, out of, during, or across subsequences. A branch that starts and ends either before or after a subsequence is permitted.
- User is responsible for avoiding mismatched loop start or end during a subsequence interval.

Avoid the following combinations in the main sequence, during a subsequence:

- Delay table inside a loop: the time for the main sequence will be incorrectly computed.
- Multiple delay tables: the time for the main sequence will be incorrectly computed.

Avoid the following in the subsequence:

- No loops or conditional branches are allowed in the subsequence.
- No half-d or waveform tables are allowed in the subsequence.
- No delay tables are allowed in the subsequence. It must have constant timing.

Frequency Hops

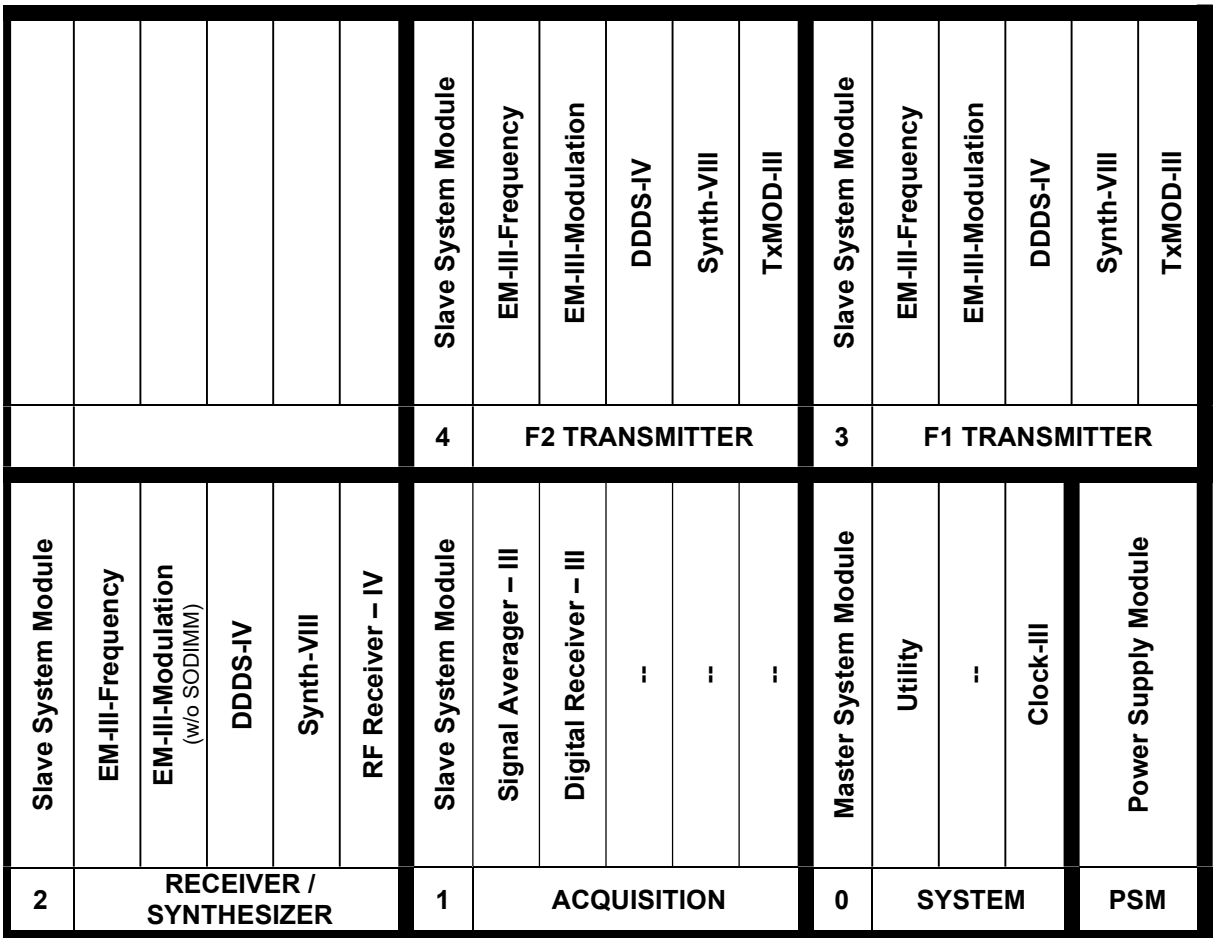
General rules:

- At least one event must intervene between frequency hop events.
- The time between hops (start of one hop event up to start of next hop event) must conform to the requirements of the synthesizer type in use.
- When considering recycle time for loops with half-d frequency tables, count each one as 6 tables (DDDS-III) or 9 tables (DDDS-II).

Special cases:

- Allow about 10 μs between frequency tables if an EL event intervenes.

Chassis Layout Diagram:



BP1 LP Outputs*

DB-25 pin	Signal	Config.con Assignment	Event Module
1	**LP20**	**Scope_Trig**	EM_Sys_0
2	LP21	**TTL_1**	
3	LP22	**TTL_2**	
4	LP23	**TTL_3**	
14	GND	GND	
15	GND	GND	
16	LP22 inv	N/A	
17	LP23 inv	N/A	
5	LP20	**TTL_4**	EM_Sys_1
6	LP21	**TTL_5**	
7	LP22	**TTL_6**	
8	LP23	**TTL_7**	
18	GND	GND	
19	GND	GND	
20	LP22 inv	N/A	
21	LP23 inv	N/A	
9	LP20		EM_Sys_2
10	LP21		
11	LP22		
12	LP23		
22	GND	GND	
23	GND	GND	
24	LP22 inv	N/A	
25	LP23 inv	N/A	
13	GND	GND	

* Pins / Signal addresses with no Config.con assignment, N/A, or GND are available for use. Contact Tecmag Support for assistance with enabling these signals.

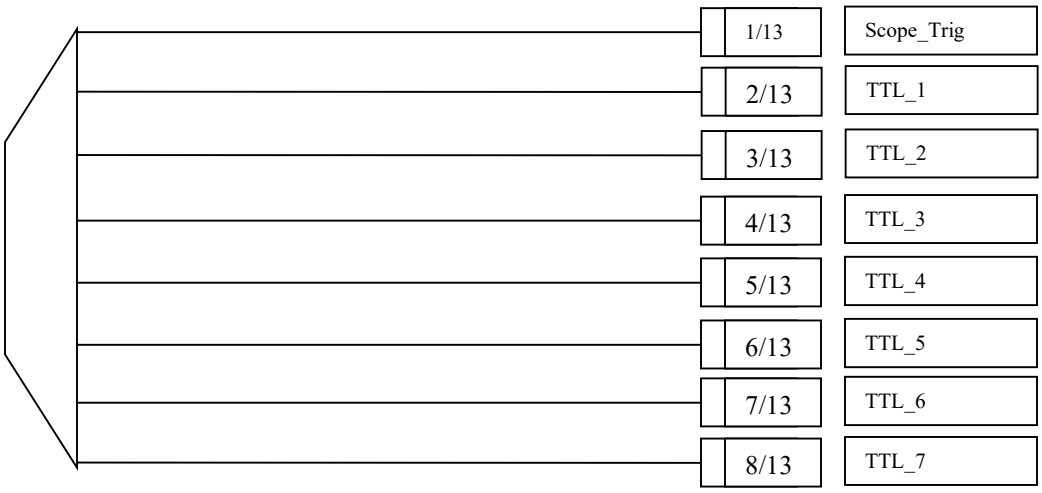
BP2 LP Outputs*

DB-25 pin	Signal	Config.con Assignment	Event Module
1	LP20	**TTL_8**	EM_Sys_3
2	LP21	**TTL_9**	
3	LP22	**TTL_10**	
4	LP23	**F1_UnBlank**	
14	GND	GND	
15	GND	GND	
16	LP22 inv	N/A	
17	LP23 inv	N/A	
5	LP20	**TTL_11**	EM_Sys_4
6	LP21	**TTL_12**	
7	LP22	**TTL_13**	
8	LP23	**F2_UnBlank**	
18	GND	GND	
19	GND	GND	
20	LP22 inv	N/A	
21	LP23 inv	N/A	
9	LP20		
10	LP21		
11	LP22		
12	LP23		
22	GND	GND	
23	GND	GND	
24	LP22 inv	N/A	
25	LP23 inv	N/A	
13	GND	GND	

* Pins / Signal addresses with no Config.con assignment, N/A, or GND are available for use. Contact Tecmag Support for assistance with enabling these signals.

DB25 LP Connectors

BP1



BP2

